"MODULAR RECYCLING STATION"

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Abstract

This project presents the design and implementation of a Modular Recycling Station (MRS) for a human mission to Jezero Crater, Mars, aimed at establishing a sustainable system for the classification, cleaning, transformation, and reuse of inorganic waste such as plastics, textiles, metals, and regolith. The MRS operates under the CCR System (Categorize, Clean, Recycle) framework, achieving up to 85% material recovery while minimizing the use of water (5 L/week) and crew time (2 h/week).

The system consists of three main modules:

CCR System, responsible for preprocessing and recycling materials using dry cleaning methods and a three-layer filtration system made of polyester, nylon, and Nomex. Ferrous metals are forged using electromagnetic induction, non-ferrous metals are reused for structural and electronic applications, thermoplastics are reshaped via compression molding, and textiles are repurposed for insulation, filters, and clothing.

Production and Logistics (P&L) System, which employs a modular mechanical press built from recycled metals to produce tools, furniture, containers, and structural components, ensuring total material reuse and energy efficiency.

Problem Solver System, designed for on-demand innovation, converting recycled materials into functional or celebratory items, and promoting experimental reuse such as polymer repolymerization and small-scale bioprocessing.

The MRS can be constructed with less than 50 kg of prefabricated equipment, ensuring self-sufficiency and operational longevity of 3–5 years. This approach demonstrates that waste can be transformed into valuable resources in extraterrestrial environments, contributing to circular sustainability and autonomy in future interplanetary missions.

Introduction

The challenge that our team selected was to design and implement a sustainable recycling and reuse system for inorganic waste during a human mission in Jezero Crater, capable of classifying, cleaning, transforming, and utilizing plastics, textiles, metals, and regolith, maximizing resource recovery while minimizing water, energy, and crew time consumption.
Our team selected the best options, the most refined processes, and the more suitable ways to proceed for the astronauts to be able to build their own workshop, bringing a reliable solution but more than that, a long-term solution to the recycling problem in an area that is way beyond the normal boundaries of recycling.

Specific Objectives

- Establish the *CCR System* (Categorize, Clean, Recycle) with standardized procedures for each material type (plastics, textiles, ferrous/non-ferrous metals, regolith).
- Achieve 85% recovery (by mass) of processable inorganic waste during the mission.
- Limit water use for cleaning processes to 5 L/week, using dry cleaning and recirculation/filtration when water use is unavoidable.
- Reduce crew time dedicated to recycling operations to 2 h/week through simplified workflows and batch processing.
- Prevent harmful emissions and byproducts by prohibiting incineration and controlling fumes and particulates.
- Standardize reusable outputs:
  - **Plastics:** thermoformed sheets or panels and composite blocks (plastic + MGS1).
  - **Textiles:** filters, covers, insulation, and protective layers.
  - **Metals:** supports/hardware and repair parts through controlled deformation or forging.
- implement quality control (visual inspection, basic tensile/rigidity measurements, and leak verification for filters) and batch traceability.
- Define preventive maintenance for equipment (booths, presses, ovens 200 °C for plastics, cutting/brushing tools).

Mission Scenario Objectives:

- **Habitat Renovations:** Reuse cubic frames and packing foams as storage modules, insulation, and panels; transform packaging plastics into sheets and blocks combined with regolith for internal bases.
- **Cosmic Celebrations:** Convert textiles and packaging materials (thermal bags, wrappers) into reusable decorations, containers, and props (banners, covers, organizers) without generating additional waste.
- **Bold Discoveries:** Reuse filters/meshes and leftover carbon as filtration media, adsorbents, and technical padding; repurposed resealable bags and nitrile gloves as laboratory consumables.

Scope and Assumptions:

- Operation within the habitat and, when applicable, in controlled exterior areas.
- Continuous electrical supply (no power generation/storage included).
- Availability of in-situ MGS-1 regolith.
- The system prioritizes immediate reuse/recycling over long-term material development.

Success Indicators (KPIs and Targets):

- Waste recovery: 85% by mass.
- Process water: 5 L/week (90% recirculated).
- Crew time: 2 h/week.
- Safety incidents: 0 events causing service interruption. Recycled part rejection rate: 10% per batch.

Our Resolution

Our resolution is based on a modular station that is able to be built on the martian surface without the need of sending more than 50 kg of equipment, which ensures the accessibility and capacity of the crew to build this station in less than a year, and having a working lifespan of at least three to five years after its built.
This also ensures that the resolution we have come to, is able to bring a safe environment, a space open enough for the crew to make modifications to the whole work space and make the reparations safe and easy for the crew to fix.

Safety Measures

- **Protective Gloves:** Nitrile gloves and gloves that are fireproof, while handling metals, plastics or any materials.
- **Facial Protection:** Safety Glasses or masks that help prevent any dust or splashes while fusion or cleaning.
- **Protective Clothing:** Antistatic Suits or textile layers (Nomex, Kevlar) in order of reducing contamination and protecting from heat and abrasion.
- **Dust Containment and Air Quality:** Perform cleaning and sanding inside sealed booths with controlled airflow.
  Minimize dust dispersion when transferring materials by using sealed containers and slow, controlled movements.
- **Water Management:** Keep water usage to a minimum and always filter before reuse.
  Avoid spills; contaminated water may contain Martian particles harmful to machinery and human health.
  Replace or clean filter layers (polyester, nylon, Nomex) regularly when saturated to maintain water quality.
- **Heat and Energy Safety:** Use controlled low-temperature heating (200 °C for plastics) to avoid toxic fumes.
  Melt and forge metals inside protected thermal enclosures with temperature monitoring.
  Allow hot pieces to cool down in isolated and marked areas to prevent accidental burns.
- **Handling Sharp or Heavy Materials:** Always check for sharp edges on cut metals or brittle plastics.
  Use cut-resistant gloves when handling ferrous and non-ferrous scrap.
  Lift heavy objects using proper techniques and mechanical support to prevent injuries, especially in low gravity.
- **Emergency Procedures:** Keep first aid kits and burn treatment materials near work areas.
  Define evacuation routes from pressurized booths in case of overheating or gas leaks.
  Continuously monitor air quality; stop work if particle levels exceed safe limits.
- **Equipment Maintenance:** Clean tools and booths frequently to prevent dust accumulation, which can damage filters and pose a fire hazard.
  Inspect and replace brushes, seals, and vacuum filters before each use.

Ensure compressed air systems are filtered and dry to avoid spreading contaminants.

Systems (Modules)

For the team to be able to establish and use correctly the *Modular Recycling Station* there are three systems (modules) that should be running correctly and at their full capacity. That is because the station is designed in different steps that depend on the usage of one module or another. At the same time, there should always be at least one person working in one module for the others to be able to work in operative conditions.

The whole modules are designed for their construction without the need of any material other than the tools that the crew already has in hand; this involves drills, knives, welders, saws, etc.
These modules are also able to be restructured, re-engineered, and reused depending on the situation that the crew finds themselves in.

1.- CCR System (Categorizing, Cleaning, and Recycling)

Categorizing.

This is a system developed for the preprocessing of materials, as a first step the garbage is classified in Plastics, Textiles and Metals.

- **Plastics:** bottles, covers, bags, packaging, gloves, wrapper, tape, bubble, wrap, air cushions, resin, etc.
  - **Thermoplastics:** Polyethylene, Polypropylene, Polivynil Chloride, Polysterene, etc.
  - **Thermosets:** Epoxy, Polyurethane, Polyester, etc.
- **Textiles:** wipes, nylon, polyester, Nomex, paper, clothes, etc.
- **Metals:** Aluminum, Stainless Steel, Copper, etc.
  - **Ferrous Metal:** Nails, Screws, Tanks, etc.
  - **Non-Ferrous Metal:** Panels, containers, Electrical Wires, Coatings, Structures, etc.

Also, we have other materials that would be catalogued as undesired residue:

- **Dust:** Plagioclase, Basalt, Pyroxene, Olivine, Mg-sulfate, Ferrihydrite, Hydrated Silica, Magnetite, Anhydrite, Fe-Carbonate, Hematite.

The next step would be cleaning the residues.

Cleaning.

This step is based entirely on the usage of brushes or rags made of plastic and textiles astronauts recover. This is because the process of cleaning is described as dry cleaning, which involves no usage of water and just removing residue and dust using brushes or rags.

They use tools used for cleaning that are already available for the astronauts.

The process of cleaning different materials varies depending on their category:

- **Plastics:** Depending on their form it's the tool that'll be used. If the plastic was exposed to any liquids or any form of organic material then a rag would be used, as if there's any residue that's non material and non sticky then a brush used.
  At the same time we use a filtered compressed air cabin that will remove any sediment or other materials that are embedded into the plastics.
- **Textiles:** They go through a shaker that removes any sediment and material, after that the textile is brushed and if the textile is hydrophilic then it's rolled with a roller and the liquid that's expelled gets collected in another bin.
  After that the textile is vacuumed to get rid of any materials.

- **Metals:** The metals go through the same filtered compressed air cabin that will remove materials, the liquids that are filtered in the process of the textiles then gets used to clean the metals with a rag, and if the metals are rusted we can use compressed air with martian dust to clean it as in sandblasting.

After the process of cleaning textiles, there will be water that has been contaminated with different non-organic materials. For reusing this water in the metal process, we would be in need of a filter that makes sure that the water is ready to be used again.

This filter is made in the next detailed process:

The filter would be made of three different layers, that being the most upper part (Thin), the middle layer (Medium) and the last layer (Thick).

For the most upper part (Thin) of the filters would need a thin layer made of a pressed fabric, ensuring that this fabric will hold the smaller particles. Then the middle layer (Medium) contains a more open fabric than the first layer that helps the filter stop any bigger particle. And finally the last layer (Thick) supports the whole structure and is the final filter, between all these layers there would be cotton added.

This filter in order would be:

- **The Most Upper Layer:** Polyester.
- **The Middle Layer:** Nylon.
- **The Last Layer:** Nomex.

For the casing of the filter, it can be made of stainless steel or any textile fabric that withstands the weight of the whole filter.

Recycling.

From the four categories that we have, for recycling we have to follow different steps and methods:

- **Metals:** For the ferrous metals; they are going to be used for tool making and for creating new workbenches or environments for more recycling methods in the future.

  These metals are going to be worked in a forge with a hydraulic press, heating the metal via electromagnetic induction using a copper coil.

  For Non Ferrous metals, they are going to be used in creating the copper coil used in electromagnetic induction for the ferrous metals, for creating structures like using aluminum, and for electronics these Non Ferrous metals will be used to produce wires, radiators, electronic welding, and other useful tools.

- **Plastics:** For thermoplastics they would be crushed into pellets of 5mm, after that they would be introduced into a mechanical press that has a heated plate and lets the plastics fusion and be pressed into a mold that was designed and made before with metal.

  With this we are able to use the thermoplastics for making tools or any other materials that could be used in the next systems.

  While making molds and injecting them with thermoplastics helps with the creation of new tools it also has the problem that with them there's toxic fumes produced, to solve this problem we have come to the solution of using Photo catalysis for turning these toxic fumes into different fumes that are less harmful, also this increases the expected usage of the extractors that will be used for taking care of the fumes that are produced by the fusion of thermoplastics.

  For Thermosets they would be crushed into pieces of less than 10mm, and mixed with resin to coat the flooring of the new habitat made with reused material.

- **Textiles:** For Textiles they will be used as rags, as insulation, as new clothes, also parts of filters and for handles of different tools.

2.- P&L System (Production and Logistics)

Utilize inorganic waste generated on Mars (plastics, textiles, metals, and EVA) by transforming it sustainably into tools, furniture, and containers useful for life and work on the Red Planet. Our workshop operates under three fundamental principles:

Total Reuse: Nothing is discarded; everything is transformed.

Energy and Water Efficiency: Water and energy use are minimized in all processes.

Modular Adaptability: Tools and machinery are adaptable, reusable, and modular.

Workshop Division.

The workshop is divided into four main zones, organized to facilitate the flow of materials:

- **Sorting Zone:** Manual separation of inorganic waste (plastics, textiles, metals).
- **Cleaning Zone:** Dry cleaning or with minimal use of recyclable liquids.
- **Compacting and Molding Zone:** Creation of compact blocks of plastic or metallic materials for reuse.

  Uses a multifunctional mechanical press built from recycled materials.
- **Production Zone:** Manufacturing of new tools, furniture, modular structures, containers, etc.
- **Modular Mechanical Press for Compacting and Molding:** A modular, low-energy mechanical press is designed, built from metallic waste (recycled aluminum and copper). This press is used both to compact plastics and to mold metals from scrap.

Figure 1: Explanatory illustration of the mechanical press

Technical Specifications.

- **Main Body:** Made with removable metal plates (ideal for changing molds or materials).
- **Compacting System:** A lid connected to a manual lever that transmits force through a central metal tube.

  When the lever is actuated, the lid descends and compacts the material inside the cube.
- **Active Thermal Base:** Made of aluminum or copper, it incorporates an induction coil for uniform heating.

  Allows reaching the necessary temperatures to deform plastics such as PLA, ABS, and PETG, without reaching their combustion point.

  Control system to adjust temperature.
- **Removable walls:** Facilitate the extraction of compacted blocks.

  In metal molding, the walls are removed to allow the pouring of molten metal into flexible molds.

- **Fume Capture and Filtration System:** During the fusion of plastic pellets (PLA, ABS, PETG) residues from 3D printing— potentially toxic vapors are generated. The following systems are implemented:

  - **Fume Capture and Filtration System:** Hermetically sealed fusion chamber..

    Upper duct that directs vapors to a thermal condenser or activated carbon filter.

    The vapor cools, condenses, and is collected in an isolated compartment for analysis or controlled disposal.

  - **Thermal Protection:** The entire area around the press is internally coated with thermal insulation materials.

    Equipment and walls are protected to prevent heat dispersion to other workshop areas.

Products that can be Created.

Thanks to the combination of heat, pressure, and modularity, the press allows the creation of a wide range of useful objects on Mars:

- **Plastic construction blocks:** used to assemble internal structures or insulate compartments.

- **Waste containers:** improving workshop organization.

- **Basic tools:** handles, levers, storage boxes.

- **Functional furniture:** work tables, modular chairs, benches.

- **3D printer parts:** gears, supports, frames.

- **Insulating panels:** made from compact plastic + recycled textile for thermal/acoustic insulation.

3-. Problem Solver System

Problem Solver.

This module is more oriented to the process of solving problems directly using the recycled material, this means that the module will always be part of the *Modular Recycling Station* even if the other modules are inactive.
With this module we can bring a solution to the two remaining objectives that were set for us:

- **Cosmic Celebrations:** For solving this objective we have divided the ideas of what the crew is able to do depending on the material that is already categorized:
  Metal: One of the ideas we've come with is using the metal for making a knife to cut the cake. Another idea is to use aluminum for making chairs or even a table for the crew to sit and enjoy the cake.
  Repolymerization of Recycled Textiles to Obtain Functional Polymeric Materials. This strategy utilizes synthetic textile waste, such as polyester and nylon, to generate new polymers with properties similar to those of virgin materials.
  The general procedure consists of the following steps:

  - Select and prepare the textiles. Identify and clean the synthetic fibers to be recycled, removing impurities and incompatible materials.

  - Controlled depolymerization or dissolution: The selected fibers undergo specific chemical processes, such as glycolysis, to break down the existing polymer chains and obtain pure monomers or chemical precursors.

  - Purification of monomers: The resulting monomers are purified using methods such as filtration, distillation, or recrystallization to remove impurities that could interfere with new polymerization.

- Repolymerization: The purified monomers undergo a controlled polymerization reaction (by condensation or addition, depending on the type of monomer) to form new polymer chains.

- Processing the Resulting Polymer: The new polymer can be molded, extruded, or transformed into filament for 3D printing. This allows for the production of customized, functional objects such as decorations, figures, and utensils for celebrations and events.

Plastic: Some ideas were to use the covers for making decorations that hang in the ceiling, other idea was to use nitrile gloves for balloon making.
We propose using the compression molding process for the treatment and reuse of thermoplastic materials.
First, the plastic waste is shredded and cleaned to obtain uniform fragments that facilitate subsequent handling. The resulting material is placed between two temperature-resistant metal plates and subjected to controlled mechanical pressure to compact the particles. Once the desired pressure is reached, the assembly is heated to a moderate temperature, which is sufficient for the polymer chains of the thermoplastics to soften and fuse through molecular reorientation without reaching the point of thermal degradation.

This process prevents the emission of toxic gases or volatile compounds.
The result is a solid sheet or molded part that can be used in various applications depending on the mold design. These products may include decorative collars, structural bases, reusable plates, or display elements.

Textiles: One idea is to use polyester and nylon, like in this detailed example:
To make a birthday hat with polyester and nylon, cut a quarter-circle pattern from polyester fabric and roll it into a cone. Seal the edges by lightly melting them and secure the seam with glue or stitching. Reinforce the inside with a nylon strip or mesh for strength, and attach an elastic or nylon string under the base to hold it on the head. Finish by decorating the hat with ribbons, paint, or small ornaments, and let everything dry before use.

Our team also came to a fun solution for another type of drinks in Mars. We will use two processes to carry out the elaboration of a homemade wine.

- Process 1: Through the use of juices or any type of sugar water, we could boil and crystallize the juice to separate the water and obtain fructose and glucose when it crystallizes. The evaporated water can be collected and reused as normal water, so nothing would be wasted.
  From this, we obtain our first raw material:

- Process 2: Now, to obtain the second raw material, we will use peels or residues from grapes, apples, or hibiscus (jamaica).
  In this case, we will reuse apple peels and cores or any available form of the fruit. We separate the fruit material before sending it for recycling or disposal.
  Then, we place it in a container filled with water kept at a temperature between 25°C and 35°C.

Once the fruit is submerged, we cover the container with a mesh or breathable lid, allowing air exchange. After 2 to 4 days, we will obtain the yeast culture needed to create a type of wine.
It should be noted that temperature control and hermetic conditions are crucial, since any variation could lead to the growth of non-edible bacteria.

If everything goes well, we will obtain yeast, which can then be separated from the liquid and stored in a Petri dish. Now that we have both raw materials yeast, fructose, and glucose we can move on to the final step, which requires equal care.

In a bottle or thermos-type container, mix water, fructose-glucose, and yeast.
Stir the mixture and leave it to ferment under conditions between 25°C and 35°C. After 24 hours, we will obtain a homemade wine with a noticeable alcohol content.

- **Bold Discoveries:** The most efficient and fastest way to take advantage of the materials from the "Bold Discoveries" challenge is simply by recycling everything.
Because when thinking and reviewing, any way to create something significant would be through making carbon fiber from cellulose. However, we would face the problem that this is a weak and inefficient type of fiber, at least under the conditions we are in.
Creating carbon fiber from polyacrylonitrile (PAN) would consume more materials than recycling, or involve a process that is too long for the astronauts to carry out efficiently and quickly.
Therefore, the only viable solution is to recycle the elements one by one using the recycling modules we already have on hand.

A list would look like this: EVA Waste could be reused due to the amount of thermoplastics in the modular press for compacting.

Instrument Mesh Filters reuse for water recycling in the CCR step. Resealable Bags and Nitrile Gloves treatment of thermoplastics and thermosets. And finally, most importantly, carbon extracted from CO2.
This carbon could be treated with oxygen, hydrogen, and nitrogen to obtain activated carbon, which can be stored for any occasion in which we want to implement this compound.

Conclusion

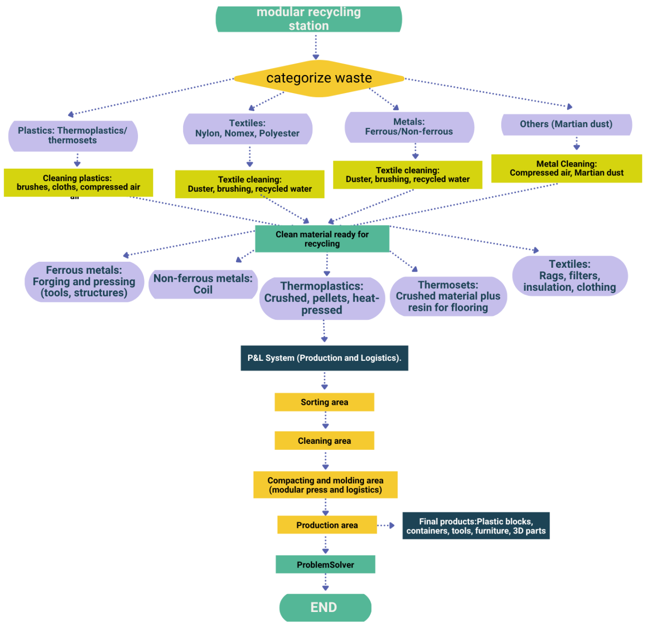

Figure 2: Flow chart on the function of the MRS

The conception and construction of our modular recycling station within the context of a human mission to Jezero Crater represented a complex process that demanded intense collaborative effort and rigorous planning. The central objective of the project was to design and implement a sustainable system for the classification, cleaning, transformation, and reuse of inorganic waste, including plastics, textiles, metals, and regolith, with the aim of maximizing resource recovery while minimizing water, energy, and crew time consumption.

The developed system was structured with standardized procedures for each type of material, achieving a recovery target of at least 85% of processable waste. Cleaning methods were established with minimal water usage, employing dry cleaning techniques and filtered liquid recirculation when water use was unavoidable. Crew time dedicated to recycling operations was reduced through simplified workflows and zone-based processing. Additionally, strict measures were implemented to control emissions and by products, avoiding processing that generates contaminants and limiting the dispersion of particles and harmful gases.

The system was organized into different fundamental zones, In the classification phase, waste was segregated into plastics (thermoplastics and thermosets), textiles, ferrous and non-ferrous metals, and regolith, while undesired materials such as dust and fine particles were cataloged as secondary waste. Material cleaning was conducted using brushes, rags, and filtered compressed-air booths, adapting methods to the properties of each material and preventing water contamination through a three-layer filtration system composed of polyester, nylon, and Nomex, with structural support and intercalated cotton.

The recycling process involved different strategies depending on the type of waste: ferrous metals were transformed into tools, supports, and structural elements through controlled forging with electromagnetic induction; nonferrous metals were used for coil manufacturing, structures, and electronic components; thermoplastics were melted into blocks and molds using a modular mechanical press with a thermal base, applying photocatalysis to neutralize toxic vapors; thermosets were crushed and mixed with resins for flooring coatings; and textiles were repurposed as rags, insulation, clothing, filters, and tool handles.

The production and logistics (P&L) system operated under three fundamental principles: total reuse of waste, energy and water efficiency, and modular adaptability of tools and machinery. The station enables the creation of plastic construction blocks, containers, basic tools, functional furniture, 3D printer parts, and insulating panels, completing a full reuse cycle within the Martian habitat.

The advantages of the system include its ability to fully reutilize available materials, reducing resource consumption and promoting sustainability; the modular structure, which facilitates maintenance and adaptation to different functions; and the promotion of creativity and collaboration among crew members by manufacturing tools from recycled materials. However, the system has certain limitations. Construction of the station was complex and required high team coordination. Some tools, being made from recycled materials, may have restrictions in strength or precision. For example, the modular mechanical press designed to compact plastics and mold metals is not automatic and may require the simultaneous operation of two people, implying considerable physical effort. Additionally, the system demands constant maintenance to ensure safe and efficient operation, including filter cleaning, tool inspection, and quality control of recycled materials.

In conclusion, the implementation of this modular recycling station has demonstrated that, despite inherent technical and logistical challenges, teamwork and optimal use of available resources allow the transformation of inorganic waste into functional and sustainable solutions. This project not only represents an advance in waste management and material reuse in extreme environments but also constitutes an example of innovation and adaptability applicable to future interplanetary missions, fostering a comprehensive approach to sustainability and efficiency in space exploration.

Also our team counts with an HTML archive, that shows the information on this article in a more simpler way.

References

[1] Linne, D. L., Palaszewski, B. A., Gokoglu, S., Gallo, C. A., Balasubramaniam, R., & Hegde, U. G..(2014) *Waste management options for long-duration space missions: When to reject, reuse, or recycle. NASA Glenn Research Center.*. AIAA SciTech.

[2] *Non-Metabolic Waste Items, Materials, and Commercial Equivalents – Foam Packaging.*

[3] Kimbrel, B (2024),*Waste Materials Recycling for In-Space Manufacturing* [Poster], Marshall Space Flight Center.

[4] University of Central Florida. (s,f), *MGS-1 Mars Global Simulant.*

[5] Howe, A. Scott; Howard, Robert, (2010), *Dual use of packaging on the moon: Logistics-2-Living.* AIAA 40th International Conference on Environmental Systems, Barcelona, Spain,

[6] NASA, (2025), *Estudio sobre por qué Marte es rojo respalda la teoría de un posible pasado habitable.* NASA Ciencia

[7] Kang, H., Washington, A., Capobianco, M. D., Yan, X., Vera Cruz, V., Weed, M., . . . Gu, J.(2023),*Concentration-Dependent Photocatalytic Upcycling of Poly(ethylene terephthalate) Plastic Waste.*ACS Materials Letters, 5(11), 3032–3041.